

5. To Apply Single Quotes, Preserve Literal Content, Ignore Variable Expansion, Store Quoted Strings, Validate Parsing Results, Test Edge Cases.

To Apply Double Quotes, Preserve Spaces, Allow Variable Expansion, Parse Nested Tokens, Validate Outputs, Test Quoted Commands

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <string.h>
#include <sys/types.h>
#include <sys/wait.h>

int main() {
    int fd[2];
    pid_t pid;
    char buffer[100];

    // Create pipe
    if (pipe(fd) == -1) {
        perror("Pipe");
        exit(1);
    }

    pid = fork();

    if (pid < 0) {
        perror("Fork");
        exit(1);
    }

    if (pid == 0) {
        // Child - Consumer
        close(fd[1]); // Close write end

        read(fd[0], buffer, sizeof(buffer));
        printf("Child received: %s\n", buffer);

        close(fd[0]);
    }
    else {
        // Parent - Producer
        close(fd[0]); // Close read end

        char message[] = "Hello from Parent using Pipe!";
        write(fd[1], message, strlen(message) + 1);

        close(fd[1]);

        write(fd[1], message, strlen(message) + 1);

        close(fd[1]);

        wait(NULL);
    }

    return 0;
}
```

```

GNU nano 2.7.1
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main() {
    int fd[2];

    if (pipe(fd) == -1) {
        perror("Pipe");
        exit(1);
    }

    pid_t pid1 = fork();

    if (pid1 == 0) {
        // First child: ls -l
        close(fd[0]); // Close read end

        dup2(fd[1], STDOUT_FILENO);
        close(fd[1]);

        execlp("ls", "ls", "-l", NULL);

        perror("execlp");
        exit(1);
    }

    pid_t pid2 = fork();

    if (pid2 == 0) {
        // Second child: grep ".c"
        close(fd[1]); // Close write end

        dup2(fd[0], STDIN_FILENO);
        close(fd[0]);

        execlp("grep", "grep", ".c", NULL);

        perror("execlp");
        exit(1);
    }
}

```

```

}

// Parent
close(fd[0]);
close(fd[1]);

wait(NULL);
wait(NULL);

return 0;
}

```

```
thumu_hemanth_kumar@Hemanths-TUF-Laptop: ~/OSPractical$ nano practical5.c
thumu_hemanth_kumar@Hemanths-TUF-Laptop: ~/OSPractical$ gcc practical5.c -o practical5
thumu_hemanth_kumar@Hemanths-TUF-Laptop: ~/OSPractical$ ./practical5
Child received: Hello from Parent using Pipe!
thumu_hemanth_kumar@Hemanths-TUF-Laptop: ~/OSPractical$ nano practical5parent.c
thumu_hemanth_kumar@Hemanths-TUF-Laptop: ~/OSPractical$ gcc practical5parent.c -o practical5parent
thumu_hemanth_kumar@Hemanths-TUF-Laptop: ~/OSPractical$ ./practical5parent
-rw-r--r-- 1 thumu_hemanth_kumar thumu_hemanth_kumar 14 Aug 8 09:48 copy.txt
-rwxr-xr-x 1 thumu_hemanth_kumar thumu_hemanth_kumar 16352 Aug 8 08:29 practical1
-rw-r--r-- 1 thumu_hemanth_kumar thumu_hemanth_kumar 606 Aug 8 08:28 practical1.c
-rwxr-xr-x 1 thumu_hemanth_kumar thumu_hemanth_kumar 16312 Aug 8 09:48 practical2
-rw-r--r-- 1 thumu_hemanth_kumar thumu_hemanth_kumar 951 Aug 8 09:48 practical2.c
-rwxr-xr-x 1 thumu_hemanth_kumar thumu_hemanth_kumar 16168 Aug 7 06:11 practical3
-rw-r--r-- 1 thumu_hemanth_kumar thumu_hemanth_kumar 1000 Aug 8 09:55 practical3.c
-rwxr-xr-x 1 thumu_hemanth_kumar thumu_hemanth_kumar 16176 Aug 7 06:48 practical4
-rw-r--r-- 1 thumu_hemanth_kumar thumu_hemanth_kumar 543 Aug 7 06:48 practical4.c
-rwxr-xr-x 1 thumu_hemanth_kumar thumu_hemanth_kumar 16384 Sep 12 08:21 practical5
-rw-r--r-- 1 thumu_hemanth_kumar thumu_hemanth_kumar 842 Sep 12 08:21 practical5.c
-rwxr-xr-x 1 thumu_hemanth_kumar thumu_hemanth_kumar 16304 Sep 12 08:23 practical5parent
-rw-r--r-- 1 thumu_hemanth_kumar thumu_hemanth_kumar 879 Sep 12 08:23 practical5parent.c
thumu_hemanth_kumar@Hemanths-TUF-Laptop: ~/OSPractical$
```